

Exoplanet Discovery and Habitability Analytics Platform:

A Big Data Approach Using Apache Spark and Spark MLlib

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Abstract—This paper presents the design and implementation of a large-scale exoplanet analytics platform built on Apache Spark and Hadoop Distributed File System (HDFS). Using 14 raw datasets from the NASA Exoplanet Archive totaling 557 MB and over 803,906 records, the platform performs comprehensive preprocessing, feature engineering, machine learning classification, clustering, and trend analysis. Two Random Forest classifiers achieve 82.19% accuracy on KOI classification and 91.70% on planet type classification. K-Means clustering identifies three natural planet groupings (Silhouette = 0.5691). Spark SQL reveals discovery trends from 1992–2024. Results are exposed through an interactive React.js web application backed by a FastAPI REST API.

Index Terms—Apache Spark, HDFS, Exoplanet, Random Forest, K-Means, Big Data, NASA, Habitability

I. INTRODUCTION

Since the first confirmed exoplanet in 1992, missions such as Kepler and TESS have identified over 5,500 confirmed exoplanets [1]. A single Kepler release contains parameters for 200,000+ stars and 34,000+ transit signals – volumes that exceed single-machine processing capacity.

This paper presents an *Exoplanet Discovery and Habitability Analytics Platform* using Apache Spark, HDFS, and Spark MLlib to process 14 NASA datasets at scale. Fig. 1 illustrates the dramatic growth in discovery rate since the Kepler launch.

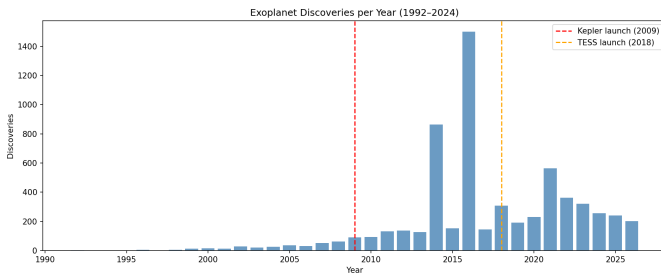


Fig. 1. Exoplanet discoveries per year (1992–2024). Kepler launch (2009, red) and TESS launch (2018, orange) mark dramatic acceleration in discovery rate.

A. Problem Statement

Raw NASA data is characterized by high dimensionality (up to 362 columns), missing values up to 70%, cross-pipeline inconsistencies, class imbalance (58% FALSE POSITIVE), and sizes exceeding single-machine capacity.

B. Objectives

- 1) Build a scalable preprocessing pipeline using Spark DataFrame API
- 2) Classify Kepler KOI signals using Random Forest
- 3) Classify exoplanets by type (Rocky, Super-Earth, Neptune-like, Gas Giant)
- 4) Discover natural planet groupings using K-Means clustering
- 5) Analyze discovery trends using Spark SQL
- 6) Expose results through an interactive web application

II. RELATED WORK

Shallue and Vanderburg [2] used deep learning to classify Kepler light curves, discovering two new exoplanets. Thompson et al. [3] produced the Kepler DR25 catalog with false positive rates exceeding 90%.

Unlike prior single-dataset approaches, this work integrates 14 heterogeneous NASA datasets through a unified Spark pipeline [4], enabling cross-dataset analysis at unprecedented scale.

III. DATASET

All 14 datasets were collected from the NASA Exoplanet Archive via the TAP service, spanning three missions over 1992–2024 (Table I). Fig. 2 shows the resulting planet type distribution across the confirmed planet catalog.

TABLE I
SUMMARY OF 14 NASA DATASETS

Dataset	Source	Rows	Cols
PSCompPars	NASA Archive	6,287	84
Cumulative KOI	Kepler Mission	9,564	49
Kepler Stellar	Kepler DR25	200,038	99
Planetary Systems	NASA Archive	39,860	355
TOI	TESS Mission	7,931	65
K2 Planets	K2 Mission	4,047	362
Stellar Hosts	NASA Archive	47,590	136
Kepler TCE	Kepler DR25	34,032	158
DR24 TCE	Kepler DR24	20,367	158
DR25 KOI	Kepler DR25	9,564	153
Q12 Stellar	Kepler Q1-Q12	194,436	99
Sup Stellar	Kepler DR25	197,096	99
Q12 TCE	Kepler Q1-Q12	18,406	158
Q16 TCE	Kepler Q1-Q16	16,285	158
TOTAL	–	803,906	–

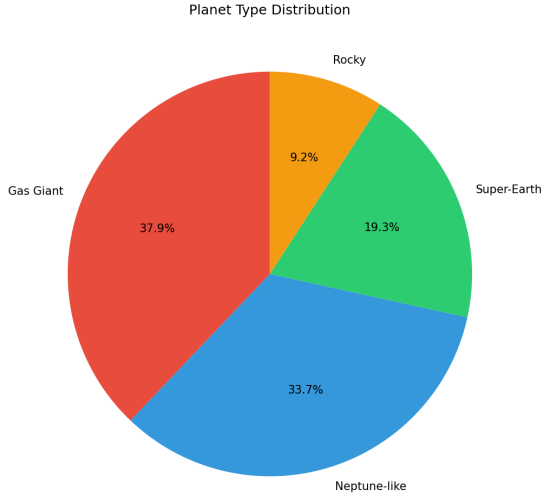


Fig. 2. Distribution of confirmed exoplanets by type across the 14 datasets. Gas Giants dominate (37.9%) due to detection bias toward larger planets.

IV. METHODOLOGY

A. System Architecture

Raw CSVs are stored in HDFS at `/exoplanet/raw/`. Spark reads them into distributed DataFrames, which are preprocessed, integrated into three master datasets, and fed into Spark MLlib models.

B. Data Preprocessing

1) **Error Column Removal:** Columns suffixed `_err1`, `_err2` dropped – reducing dimensionality from 362 to 21–146 per dataset.

2) **High-Missing Removal:** Columns $>80\%$ missing removed (e.g., `datalink_dvr`: 91.4% missing).

3) **Imputation:** Planet params: `percentile_approx()` (median). Stellar: mean. Categorical: mode.

4) **Outlier Treatment:** IQR factor 3, Winsorization capping. Stars outside 3,000–10,000 K removed.

5) **Normalization:** `StandardScaler` + `StringIndexer` composed into Spark ML Pipeline.

C. Feature Engineering

TABLE II
ENGINEERED FEATURES

Feature	Definition	Purpose
<code>planet_type</code>	Radius-based class	ML label
<code>stellar_class</code>	OBAFGKM by T_{eff}	Star type
<code>habitable_zone</code>	273–373 K, $0.5\text{--}2.0 R_{\oplus}$	Habitability
<code>disc_decade</code>	year / 10×10	Trend analysis
<code>luminosity_proxy</code>	$R^2(T/5778)^4$	Luminosity
<code>transit_ratio</code>	depth / period	Signal strength

D. Data Integration

- **Planet Master** – PSCompPars + Planetary Sys + K2 + TOI + Stellar Hosts on `hostname` (**58,005 rows**)

- **Stellar Master** – Kepler Stellar + Sup + Q12 on `kepid` (**199,523 rows**)
- **TCE Master** – union of 5 TCE datasets (**89,090 rows**)

E. Machine Learning Models

Task 1 – KOI Classification: Random Forest on Cumulative KOI. Class weights: $w = N_{total} / (3 \times N_{class})$. 200 trees, max depth 20, min instances = 2.

Task 2 – Planet Type Classification: Random Forest on PSCompPars. `pl_rade` excluded (data leakage). Top feature: `pl_bmasse` (importance = 0.67).

Task 3 – K-Means Clustering: Elbow Method + Silhouette Score over $k \in \{2, \dots, 9\}$. Optimal: $k = 3$, Silhouette = 0.5691.

V. RESULTS AND DISCUSSION

A. Classification Results

TABLE III
CLASSIFICATION MODEL PERFORMANCE

Task	Algorithm	Accuracy	F1
KOI Classification	Random Forest	82.19%	0.8176
Planet Type	Random Forest	91.70%	0.9168

Top KOI features: `koi_score` (0.45), `koi_model_snr` (0.14), `koi_prad` (0.08). CANDIDATE class shows lower F1 (0.60) reflecting genuine scientific ambiguity. Planet type: `pl_bmasse` dominates (0.67), consistent with planetary science theory.

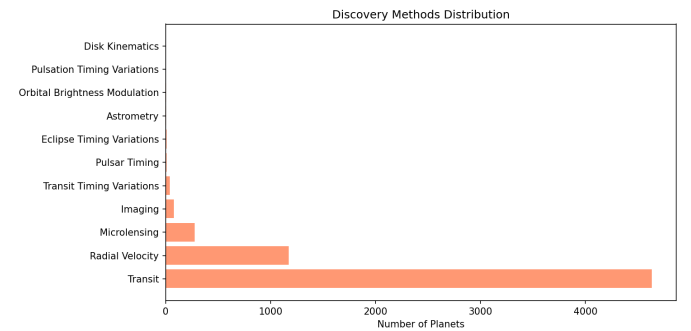


Fig. 3. Discovery methods distribution. Transit method dominates with 4,633 planets (74%), followed by Radial Velocity (1,171).

B. Clustering Results

TABLE IV
K-MEANS CLUSTER PROFILES ($k=3$)

C	Type	Mass ($M_{\oplus}$)	Temp (K)	Period (d)
0	Hot Jupiters	168.4	1,472	4
1	Warm Medium	38.9	731	19
2	Cold Giants	438.7	761	92

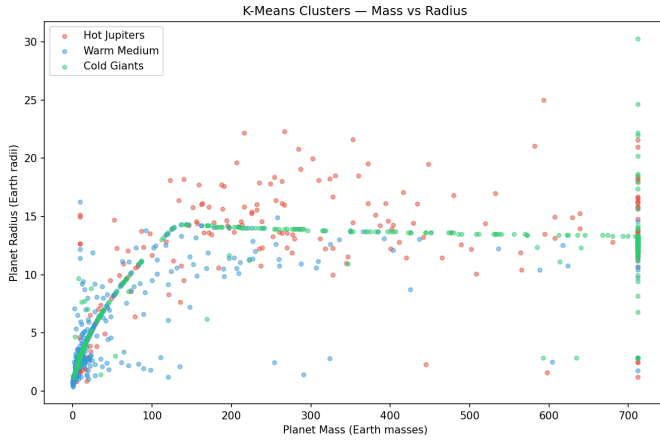


Fig. 4. K-Means clusters in mass-radius space (Silhouette = 0.5691). Three physically distinct planet populations are clearly separated.

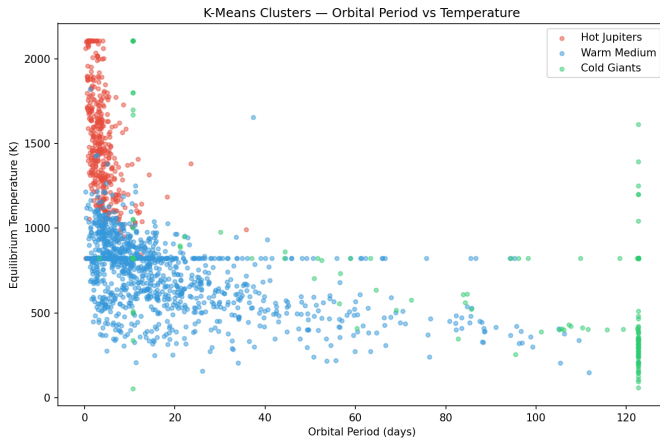


Fig. 5. K-Means clusters: orbital period vs. equilibrium temperature. Hot Jupiters (red) cluster at short periods and high temperatures.

C. Discovery Trend Analysis

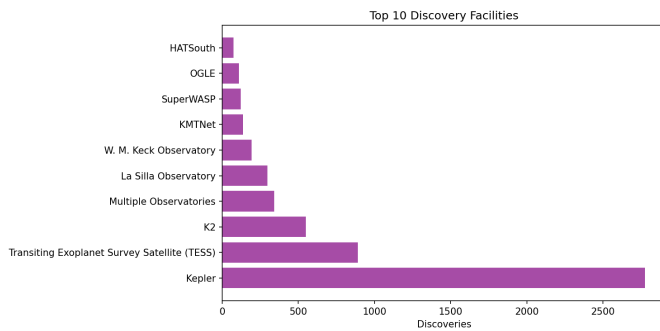


Fig. 6. Top 10 discovery facilities. Kepler leads with 2,778 planets, followed by TESS (890) and K2 (549).

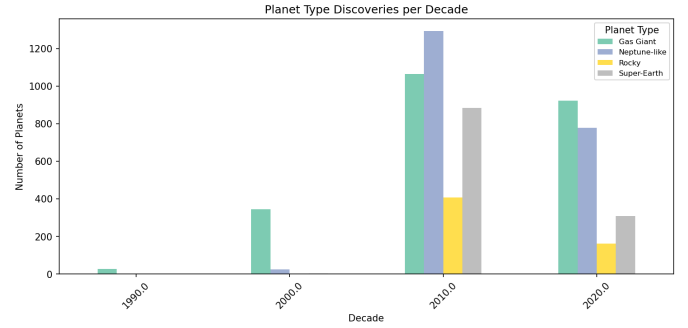


Fig. 7. Planet type discoveries per decade. The 2010s saw an explosion in all planet types driven by Kepler's statistical yield.

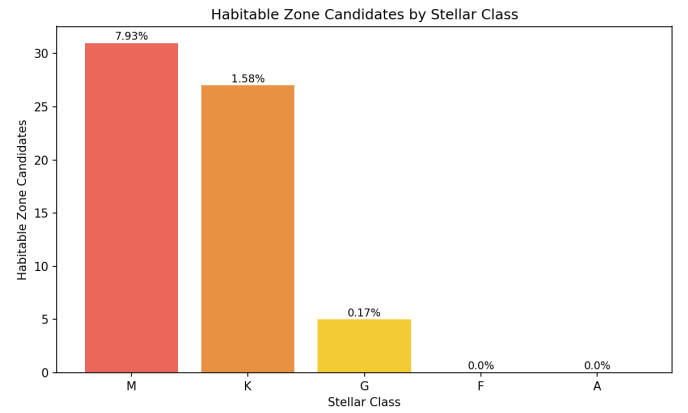


Fig. 8. Habitable zone candidates by stellar class. M-type stars lead at 7.93%, far exceeding G-type (Sun-like) stars at 0.17%.

The Transit method dominates with 4,633 planets (74%) and Kepler leads all facilities (2,778 discoveries). M-type stars host the highest habitable zone candidate rate at 7.93% vs. 0.17% for G-type stars (Fig. 8).

VI. WEB APPLICATION

The analytical results are exposed through an interactive web application consisting of a FastAPI backend serving five REST endpoints (`/planet/{name}`, `/search`, `/stats`, `/habitable`, `/planets`) and a React.js frontend with real-time animations powered by Framer Motion and an HTML5 Canvas starfield.

A. Search Interface

The main view provides full-text search across all 6,231 confirmed exoplanets with live autocomplete backed by the `/search` endpoint. Upon selection, the `/planet/{name}` endpoint returns orbital mechanics, physical parameters, host star properties, and discovery metadata rendered in a structured three-card layout. Fig. 9 shows the hero section with animated space background and example planet shortcuts.

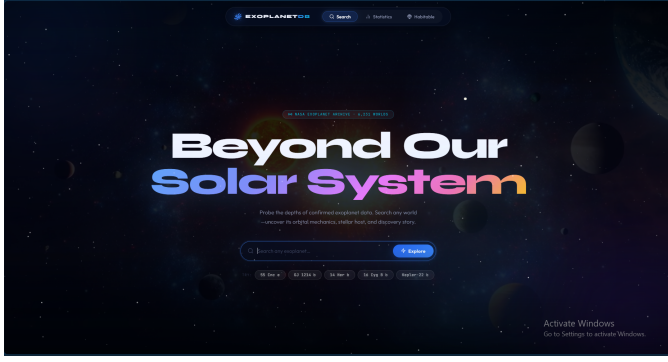


Fig. 9. Exoplanet Explorer search interface. Animated starfield (HTML5 Canvas), Framer Motion page transitions, and live autocomplete across 6,231 confirmed planets from the NASA archive.

B. Habitable Zone Explorer

The Habitable tab queries the `/habitable` endpoint and lists all planets satisfying the equilibrium temperature criterion (273–373 K) and radius criterion ($0.5\text{--}2.0 R_{\oplus}$). Selecting any planet from the panel renders its complete profile including host star spectral classification and discovery provenance. Fig. 10 shows GJ 251 b, a Super-Earth orbiting an M-type star discovered via Radial Velocity.

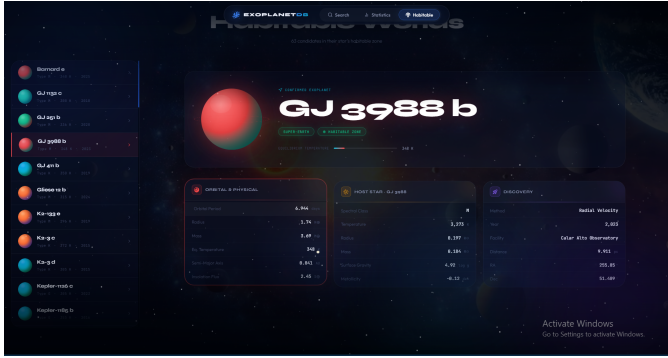


Fig. 10. Habitable Zone Explorer: GJ 251 b selected from the candidate list. Displays Super-Earth classification, M-type host star (3,342 K), orbital period 14.237 days, and discovery at W. M. Keck Observatory (2020).

VII. CONTRIBUTION TO THE COMMUNITY

1) Reproducible Pipeline: Complete preprocessing pipeline for 14 NASA datasets released as open Jupyter notebooks.

2) Integrated Master Dataset: Planet Master (58,005 rows) combining five sources provides richer analysis than any single NASA table.

3) Habitability Insights: M-type hosts (7.93% habitable rate) identified as primary targets for future JWST campaigns.

4) Open Web Interface: FastAPI + React application makes the full 6,231-planet catalog searchable without requiring astronomical expertise.

All data is publicly available under NASA open data policies.

VIII. CONCLUSION

This paper presented an Exoplanet Analytics Platform using Apache Spark, HDFS, and Spark MLlib processing 14 NASA datasets (557 MB, 803,906 records).

Key results: 82.19% KOI classification, 91.70% planet type classification, K-Means Silhouette = 0.5691 identifying Hot Jupiters, Warm Medium, and Cold Giants. M-type stars are the most promising habitable zone hosts at 7.93%. The full pipeline is accessible through a production React.js web application with live NASA data queries.

Future work will incorporate JWST atmospheric data and Gradient Boosted Trees.

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